ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba [ANSWER KEY]

Unit 2.01: Modular Arithmetic, Divisibility, Primes & Euler Totient

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 8x + 28 in the domain of Modular Arithmetic, Primes & Divisibility. By completing the square f_1x = x - h^2 + k , find the sum k + f_16 .

[ ] A: 32
[x] B: 28
[ ] C: 16
[ ] D: 26

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Complete the square: x^2 - 8x + 28 = x - 4^2 - 16 + 28 = x - 4^2 + 12 , giving minimum value k = 12 . **[M1]** 2. Evaluate at x = 6 : f_16 = 2^2 + 12 = 16 . **[M1]** 3. Sum the two quantities: k + f_16 = 12 + 16 = 28 . **[A1]** - **Final Verified Answer:** 28 .

Q2. A mathematical model in Modular Arithmetic, Primes & Divisibility is defined by the cubic function y_2x = 3x^3 - 4x^2 + 15x - 9 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 96
[ ] B: 57
[x] C: 72
[ ] D: 34

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 9x^2 - 8x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 99 - 83 + 15 = 72 . **[M1][A1]** - **Final Verified Answer:** 72 .

Q3. In Modular Arithmetic, Primes & Divisibility, two displacement vectors are given by mathbf{u}_{3} = langle 4, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: -20
[ ] B: 80
[ ] C: 172
[x] D: 60

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 49 + -87 = 36 + -56 = -20 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 4^2 + -8^2 = 80 . **[M1]** 3. Combine: -20 + 80 = 60 . **[A1]** - **Final Verified Answer:** 60 .

Q4. The linear growth profile shown in the diagram for Modular Arithmetic, Primes & Divisibility follows an arithmetic progression with first term a_1 = 5 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[x] A: 705
[ ] B: 89
[ ] C: 795
[ ] D: 1335

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 5, d = 6 : S_{15} = frac{15}{2}[10 + 14 times 6] = frac{15}{2} times 94 = 705 . **[M1][A1]** - **Final Verified Answer:** 705 .

Q5. In Modular Arithmetic, Primes & Divisibility, evaluate the exact definite integral $I_5 = \int_0^{2} 18x^2 + 16x\,\mathrm{d}x$.

[ ] A: 104

[x] B: 80

[ ] C: 96

[ ] D: 41

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Integrate term by term: $\int 18x^2 + 16x\,\mathrm{d}x = \left[6x^3 + 8x^2\right]_0^{2}$. **[M1]** 2. Substitute the upper and lower limits: $62^3 + 82^2 - 0 = 48 + 32 = 80$. **[M1][A1]** - **Final Verified Answer:** 80 .

Q6. In Modular Arithmetic, Primes & Divisibility, consider the quadratic function $f_6x = x^2 - 10x + 109$. Express f_6x in the completed-square form $x - h^2 + k$ and compute the exact value of $k + f_{612}$.

[ ] A: 222

[ ] B: 133

[x] C: 217

[ ] D: 210

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Complete the square: $x^2 - 10x + 109 = x - 5^2 - 25 + 109 = x - 5^2 + 84$, giving minimum value $k = 84$. **[M1]** 2. Evaluate at $x = 12$: $f_{612} = 7^2 + 84 = 133$. **[M1]** 3. Sum the two quantities: $k + f_{612} = 84 + 133 = 217$. **[A1]** - **Final Verified Answer:** 217 .

Q7. The graph illustrates the cubic model $y_7x = 3x^3 - 5x^2 + 33x - 16$ studied in Modular Arithmetic, Primes & Divisibility. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

[ ] A: 609

[ ] B: 496

[ ] C: 95

[x] D: 529

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Differentiate term by term using the power rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = 9x^2 - 10x + 33$. **[M1]** 2. Substitute $x = 8$: $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=8} = 964 - 108 + 33 = 529$. **[M1][A1]** - **Final Verified Answer:** 529 .

Q8. In Modular Arithmetic, Primes & Divisibility, two displacement vectors are given by $\mathbf{u}_{8} = \langle 5, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + |\mathbf{u}_{8}|^2$.

[x] A: 79

[ ] B: 5

[ ] C: 74

[ ] D: 149

Solution: **Mark Scheme & Examiner Solution Modular Arithmetic, Primes & Divisibility:** 1. Compute the inner product: $\mathbf{u}_{8} \cdot \mathbf{v}_{8} = 58 + -75 = 40 + -35 = 5$. **[M1]** 2. Compute the squared norm: $|\mathbf{u}_{8}|^2 = 5^2 + -7^2 = 74$. **[M1]** 3. Combine: $5 + 74 = 79$. **[A1]** - **Final Verified Answer:** 79 .